

Md Mujahid Hossain

GRADUATE ENGINEER · ELECTROMAGNETICS, RF & SENSING SYSTEMS

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Research Interests

Metamaterials & Electromagnetic Sensing; RF & Microwave Engineering; Antenna Design; Gas and Biosensing using Metamaterial Structures; Computational Electromagnetics; Nanoelectronics & Semiconductor Device Physics; Memristive Devices; Neuromorphic Computing & AI Hardware; Nanoscale Characterization (TEM, SEM); AI-Assisted Design and Optimization; Analog/Mixed-Signal Interfaces; IoT-Based Smart Sensing; 5G/6G and ISAC.

Education

International Islamic University Chittagong

Chattogram, Bangladesh

BACHELOR OF SCIENCE (BSc) ELECTRONIC AND TELECOMMUNICATION ENGINEERING

April 2020 - April 2025

- Thesis: Dual-Band Metamaterial Absorber for C & X Band Applications
- Supervisor: Dr. Saif Hannan
- CGPA: 3.569/4.00

Professional Experience

- JUL 2026 **R&D Intern - IT**, Azneo Limited, Chattogram
- 2024 **Graduate Teaching Assistant**, Dept. of ETE, IIUC
- 2024 **Industrial Trainee (Telecommunications & Network Operations)**, Robi Axiata Ltd., Dhaka — Network monitoring (NMC/SOC), RF systems, microwave & optical transmission
- 2025 **Undergraduate Researcher**, Dept. of ETE, IIUC
- 2023–2024 **Chairperson, IEEE Communications Society SBC**, IEEE IIUC Student Branch
- 2023–2024 **Vice Chairperson**, IEEE IIUC Student Branch
- 2023–2024 **Founding Chairperson**, IIUC ETE Career Forum
- 2023–2024 **Secretary (Technical)**, ETE Research Society, IIUC

Technical Skills

Electromagnetic Simulation: CST Studio Suite, ANSYS HFSS, ADS

Embedded & Hardware: Arduino (ATmega328P), sensor & I/O interfacing, serial communication (UART, I2C, SPI), soldering & hardware assembly

PCB & EDA: EasyEDA (schematic capture & PCB layout), Proteus (circuit simulation & debugging)

Electronics: Op-Amps, BJTs, FETs, Relays, Triacs; analog & digital circuit design

Programming: Python, MATLAB, C/C++ (embedded), Java

Lab Tools: Oscilloscope, Multimeter, DC Power Supply, Function Generator

Data & Analysis: MS Excel, Data Visualization

Core Areas: Electromagnetics, RF & Microwave Engineering, Antenna Design, Metamaterials, Machine Learning, IoT Systems

Awards & Achievements

- 2024 **Best Paper Award**, ICISSET 2024
- 2025 **Best Paper Award**, AIEE 2025
- 2024 **Finalist**, Nano Satellite Expedition, IEEE AEES BRAC University
- 2023 **Finalist**, CUET Research Fair, Chittagong University of Engineering & Technology
- 2022 **Champion**, Inter-Department Debate Competition, IIUC

Publications

JOURNAL ARTICLES

M. M. Hossain, S. Hannan, Md. I. Haque, M. A. Alawad, A. M. Alenezi, and M. T. Islam, "A Highly Sensitive Multi-Resonant Metamaterial Absorber for Wideband Hydrocarbon and Gas Detection," *Applied Physics A*, vol. 132, no. 5, p. 395, 2026. DOI: 10.1007/s00339-026-09512-3

Nasif, M.A., Shahriyar, F., Hossain, **M.M.** et al. Design and analysis of a photonic crystal fiber sensor for highly toxic gases detection. *Eur. Phys. J. Plus* 140, 22 (2025). <https://doi.org/10.1140/epjp/s13360-025-05967-0>

CONFERENCE PAPERS

Hossain, Md Mujahid, Saif Hannan, Md Ismail Hossen, Mohammed Jashim Uddin, and Mohammad Mohi Uddin. "Dual-Band Metamaterial Absorbers for C and X Band Applications to Optimize eMBB in the 5G n79 Band at 5 GHz." In 2025 International Conference on Electrical, Computer and Communication Engineering (ECCE), pp. 1-6. IEEE, 2025.

Hossen, Md Ismail, Saif Hannan, **Md Mujahid Hossain**, Norbahiah Misran, and Mohammad Tariqul Islam. "Square-Enclosed Multiple Split-Ring Resonator Metamaterial Absorber Designed for X, Ku, and K Band Applications." In International Conference on Materials and Manufacturing Technologies, pp. 147-157. Singapore: Springer Nature Singapore, 2025.

Hossen, Md Ismail, Apurba Ray Chowdhury, Saif Hannan, **Md Mujahid Hossain**, and Md Abdul Rakib Bhuiyan. "Circular split SRR cornered by staircase shaped metamaterial absorber for 5G and X band applications." In 2024 3rd International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE), pp. 1-6. IEEE, 2024.

Sifat, Md Zahidul Islam, Muhammad Mostafa Amir Faisal, **Md Mujahid Hossain**, and Md Aminul Islam. "An Improved Positioning System For 6G Cellular Network." In 2023 26th International Conference on Computer and Information Technology (ICCIT), pp. 1-6. IEEE, 2023.

Chowdhury, Shubho Ray, Apurba Ray Chowdhury, Saif Hannan, Md Mohsin Khan, Debesh Barua Samudra, Sayed Mahamud, and **Md Mujahid Hossain**. "S-Shape Polarization Insensitive Metamaterial Absorber for SAR Reduction at 3.5 GHz." In 2024 International Conference on Innovations in Science, Engineering and Technology (ICISSET), pp. 1-5. IEEE, 2024.

Chowdhury, Apurba Ray, Saifuddin Khaled Nabil, Saif Hannan, **Md Mujahid Hossain**, Md Farhan Alam, and Md Ibrahim. "Back to Back C Shaped Rotational Symmetric Metamaterial Absorber for Triple Band Applications." In 2024 International Conference on Innovations in Science, Engineering and Technology (ICISSET), pp. 1-5. IEEE, 2024.

Arfin, Md Shahadat, Md Ismail Hossen, **Md Mujahid Hossain**, and Saif Hannan. "Lander Vector Icon Embedded Metamaterial Absorber for 5G and C Band Applications." In 2024 27th International Conference on Computer and Information Technology (ICCIT), pp. 2398-2403. IEEE, 2024.

Teaching Experience

	Microcontroller, Microprocessor & Peripherals , Teaching Assistant, Dept. of ETE, IIUC —
Spr 2024	Supported lab sessions on embedded C programming, peripheral interfacing, I/O control, and hardware debugging
Aut 2024	Digital Communication , Teaching Assistant, Dept. of ETE, IIUC
Spr & Aut 2024	Analog Communication (Sessional) , Teaching Assistant, Dept. of ETE, IIUC

References

Dr. Saif Hannan

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Noakhali Science and Technology University (NSTU), Bangladesh
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Dr. Engr. Abdul Gafur

Associate Professor, Department of Electronic and Telecommunication Engineering
International Islamic University Chittagong (IIUC), Bangladesh
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